

What are Streams in Snowflake?

A **Stream** in Snowflake is an object used to **capture and track data changes (DML operations)** made to a table.

Streams record changes such as:

- INSERT
- UPDATE
- DELETE

Streams also store **metadata about these changes**, which can be used to build **data pipelines and incremental processing workflows**.

Important point:

Streams **do not store actual table data**.

They only store **change tracking information (offsets)**.

Key Concept of Streams

When a stream is created on a table:

- 1 Snowflake takes a **point-in-time snapshot of the table**
- 2 This snapshot is called an **offset**
- 3 The stream tracks all DML changes **after this offset**

This allows Snowflake to identify **new or changed data** since the stream was created.

Why Streams are Used

Streams are mainly used for **Change Data Capture (CDC)**.

This allows systems to process **only new or changed data instead of scanning entire tables**.

Common use cases:

- Incremental data pipelines
- Real-time data processing
- ETL / ELT workflows
- Data synchronization
- Audit tracking

Streams are commonly used together with **Snowflake Tasks for automation**.

Types of Streams in Snowflake

Snowflake provides **three types of streams**.

1 Standard Stream

Tracks **all DML changes** in a table.

Tracked operations:

- INSERT
- UPDATE
- DELETE

This is the **most commonly used stream type**.

2 Append-Only Stream

Tracks **only inserted rows**.

It does not track:

- Updates
- Deletes

Used in scenarios where **data is only appended**.

Example:

Event logs or transaction records.

3 Insert-Only Stream

Tracks **only inserted rows on external tables**.

Important point:

Insert-only streams are **supported only for external tables**.

Metadata Columns in Streams

Each stream automatically includes **hidden metadata columns**.

1 METADATA\$ACTION

Indicates the **type of DML operation**.

Possible values:

- INSERT
 - DELETE
-

2 METADATA\$ISUPDATE

Indicates whether the operation was part of an **UPDATE statement**.

Values:

TRUE

FALSE

3 METADATA\$ROW_ID

A **unique identifier for each row**.

This ID is **immutable** and helps track row changes.

Example 1 – Insert then Update

Step 1

Create stream on table

Stream state:

Empty

Step 2

Insert a row in the table

Stream captures:

ROW_ID	ACTION	ISUPDATE
#1234	INSERT	FALSE

Step 3

Update the same row

Stream still shows:

ROW_ID	ACTION	ISUPDATE
#1234	INSERT	FALSE

Reason:

From the stream's perspective, the row **did not exist when the stream was created**.

So the update is treated as a **new row insertion**.

Example 2 – Update Existing Row

Step 1

Insert row before creating stream

Step 2

Create stream

Step 3

Update the row

Stream result:

ROW_ID ACTION ISUPDATE

#1234 INSERT TRUE

Reason:

The row already existed before the stream was created.

So modification is treated as **update operation**.

Example 3 – Insert then Delete

Step 1

Create stream

Step 2

Insert row

Step 3

Delete same row

Stream result:

Empty

Reason:

Final table state is **same as initial state**.

So Snowflake does not record any change.

Example 4 – Delete Existing Row

Step 1

Insert row before creating stream

Step 2

Create stream

Step 3

Delete row

Stream result:

ROW_ID ACTION ISUPDATE

#1234 DELETE FALSE

Reason:

Table now has **one row less compared to stream creation time.**

Important Behavior of Streams

Streams store **only the latest DML action per row.**

If multiple operations occur on the same row, the stream reflects **the final change.**

Key Characteristics of Streams

- Streams capture **table changes**
 - Streams do **not store table data**
 - Streams store **offsets and metadata**
 - Streams support **exactly-once processing**
 - Multiple streams can be created for the same table
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Important Limitations

Streams **do not work with views.**

They also **do not work with materialized views.**

Streams work only on:

- Tables
 - External tables
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Multiple Streams on Same Table

Snowflake allows **multiple streams on the same table.**

Example:

Stream1 → used by ETL pipeline

Stream2 → used by analytics job

Each stream tracks its **own offset independently.**

Typical Interview Questions

Question

Which object in Snowflake stores DML changes and metadata?

Answer:

Table Streams

Question

Do streams store table data?

Answer:

No

Streams only store **offsets and metadata**.

Question

Which columns are created automatically in streams?

Answer:

- METADATA\$ACTION
 - METADATA\$ISUPDATE
 - METADATA\$ROW_ID
-

Question

Types of streams?

Answer:

- Standard
 - Append-only
 - Insert-only
-

Question

Can multiple streams be created for the same table?

Answer:

Yes

Quick Summary

Streams in Snowflake provide **Change Data Capture (CDC)** by tracking DML operations on tables.

Instead of processing entire tables repeatedly, streams allow systems to process **only new or modified data**, enabling efficient incremental data pipelines.

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